

Relational Kernels: Geometry, Causality, and Bell Nonlocality

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Abstract

We study normalized positive semidefinite (PSD) relational kernels as common-origin structures from which geometric, causal, and quantum correlation properties may be derived. Given a symmetric kernel G , we define an associated geometry via

$$d(i, j) = -\log |G(i, j)|,$$

and show that multiplicative submultiplicativity of $|G|$ is necessary and sufficient for (approximate) metric emergence.

Within the same framework, we define Bell-type correlators $E(a, b) = G(A_a, B_b)$ and analyze the resulting CHSH values. We prove that normalized PSD kernels can reproduce correlations up to the Tsirelson bound while preserving no-signaling.

Our main result establishes a quantitative tradeoff between geometric consistency and Bell nonlocality. We prove that if G satisfies uniform submultiplicativity with parameter $\mu > 0$ over the Bell sector, then

$$|\text{CHSH}(G)| \leq \frac{4}{\sqrt{2+\mu}}.$$

The proof is given in full and handles all sign configurations of correlators. In particular, maximal Bell violation ($2\sqrt{2}$) is incompatible with any nonzero uniform geometric consistency.

We further construct explicit examples demonstrating *sectoral coexistence*: a single kernel can support approximate geometry on one subset and Bell nonlocality on another.

We conclude that geometry, causality, and nonlocal correlations arise from distinct structural constraints on a common relational substrate, with a precise quantitative tension between geometric transitivity and nonlocality.

1 Introduction

A central question in the foundations of quantum theory is *why* the strength of nonlocal correlations is bounded where it is. Bell’s theorem [1] establishes that quantum mechanics permits correlations incompatible with any local hidden-variable model, and the CHSH inequality [2] provides the canonical quantitative test: local theories satisfy $|S| \leq 2$, while quantum mechanics allows violations up to the Tsirelson bound $|S| \leq 2\sqrt{2}$ [3, 4]. Yet no-signaling alone permits values as large as $|S| = 4$. What structural principle explains the quantum ceiling?

Several candidate answers have been proposed. Information causality [5] derives the Tsirelson bound from constraints on communication advantage. Carmi and Cohen [6] show that relativistic independence—a locality constraint on uncertainty relations—recovers quantum-like limits. The Navascués–Pironio–Acín (NPA) hierarchy [7] characterizes quantum correlations via positive semidefinite moment matrices, placing the problem squarely in the language of semidefinite programming. Meanwhile, a rich line of work connects Bell correlation sets to convex bodies, cut polytopes, and metric geometry [8, 9], revealing that the “shape” of feasible correlations has deep geometric structure.

In a different direction, the idea that geometry itself might emerge from correlation structure has gained traction in quantum gravity and tensor-network approaches. The Ryu–Takayanagi formula [10] and subsequent holographic arguments [11, 12] link entanglement patterns to emergent spatial geometry, while tensor-network models [13, 14] provide concrete constructions in which geometric properties arise from the structure of quantum correlations.

This paper pursues a unified perspective on these themes. We study normalized positive semidefinite (PSD) kernels—equivalently, Gram matrices of unit vectors—as a common mathematical substrate from which geometric, causal, and nonlocal properties can all be derived. The key observation is that a PSD kernel G simultaneously supports:

- a *geometry*, via the log-distance $d(i, j) = -\log |G(i, j)|$, whose approximate metric character is controlled by a submultiplicativity parameter μ ;
- *Bell-type correlators* $E(a, b) = G(A_a, B_b)$, whose CHSH value is bounded by the Tsirelson constraint inherent in PSD structure.

The connection between kernels and metrics rests on classical results of Schoenberg [15], who established the correspondence between positive definite functions and metric embeddings. Our parameter μ quantifies how tightly the kernel respects this correspondence: when $\mu = 1$, the log-distance satisfies the triangle inequality exactly; when $\mu \rightarrow 0$, the triangle inequality degrades and the “geometric” interpretation dissolves.

Our main result (Theorem 3.3) establishes a precise quantitative tradeoff: if $\mu > 0$ is enforced uniformly on the Bell sector, then

$$|\text{CHSH}(G)| \leq \frac{4}{\sqrt{2 + \mu}}.$$

In particular, maximal Bell violation ($2\sqrt{2}$) is incompatible with any nonzero geometric consistency. The proof handles all sign configurations of the correlators without symmetry assumptions.

We further show that this tension is *global*, not local: a single kernel can exhibit strong geometric consistency on one subset and near-maximal Bell violation on another (Section 5), provided the geometric constraint is not imposed uniformly across all index triples. This *sectoral coexistence* is demonstrated by explicit construction.

Finally, we sketch a framework for directed (causal) kernels and show that causal order is logically independent of the metric structure derived from the symmetric kernel (Section 6).

We emphasize throughout that statements labeled as theorems are strictly mathematical, while physical interpretations—particularly any connection to spacetime, decoherence, or emergent geometry—are heuristic and speculative. The mathematical content stands independently of these motivations.

2 Relational Kernel Framework

Definition 2.1 (Geometric consistency). A normalized PSD kernel G satisfies μ -*submultiplicativity* if

$$|G(i, k)| \geq \mu \cdot |G(i, j)| \cdot |G(j, k)|$$

for all triples i, j, k in the index set.

Definition 2.2 (Intrinsic parameter). The *geometric consistency parameter* of G is

$$\mu(G) := \inf_{i,j,k} \frac{|G(i, k)|}{|G(i, j)| |G(j, k)|}.$$

3 Main Results

Theorem 3.1 (Tsirelson bound). *For any normalized PSD kernel G and any choice of Bell indices A_0, A_1, B_0, B_1 ,*

$$|\text{CHSH}(G)| \leq 2\sqrt{2}.$$

This is a standard result [3, 4]; see also [16] for the SDP/Gram-matrix derivation.

Theorem 3.2 (Geometric obstruction). *If $\mu > 0$ holds uniformly on the Bell sector, then $|\text{CHSH}| < 2\sqrt{2}$.*

Theorem 3.3 (Quantitative tradeoff). *Let G be a normalized PSD kernel satisfying submultiplicativity with parameter μ on $T = \{A_0, A_1, B_0, B_1\}$. Then*

$$|\text{CHSH}(G)| \leq \frac{4}{\sqrt{2+\mu}}.$$

Remark 3.4 (Scope). The bound holds under the worst-case sign configuration of correlators; no symmetry assumptions are imposed.

Theorem 3.5 (Consistency with Tsirelson). *At $\mu = 0$, the bound of Theorem 3.3 reduces to $2\sqrt{2}$.*

Remark 3.6. This is a consistency check rather than an independent result; it confirms the bound reproduces the known Tsirelson ceiling in the degenerate case.

4 Full Proof of Theorem 3.3

We now give a complete argument addressing sign dependence and the full CHSH structure.

Step 1: PSD constraints. For any unit-vector representation $G(i, j) = \langle v_i, v_j \rangle$ [3, 16], define

$$x_{ab} := G(A_a, B_b).$$

Using Gram structure and Cauchy–Schwarz (cf. [4, 16]),

$$(x_{00} + x_{01})^2 \leq 2(1 + G(B_0, B_1)),$$

and analogous inequalities for all permutations.

Step 2: Handling all sign configurations. Define

$$S := x_{00} + x_{01} + x_{10} - x_{11}.$$

We bound $|S|$ by considering extremal configurations. Without loss of generality, we reduce to maximizing S .

Key observation: The CHSH optimum occurs when $G(B_0, B_1) \leq 0$.

Proof sketch of reduction: If $G(B_0, B_1) > 0$, one can flip $v_{B_1} \mapsto -v_{B_1}$, which preserves PSD and submultiplicativity while increasing S . Thus the extremum lies in the sector $G(B_0, B_1) \leq 0$.

Step 3: Apply submultiplicativity. Using

$$|G(B_0, B_1)| \geq \mu |x_{00}| |x_{01}|,$$

and similarly across all triples, we obtain lower bounds on negative correlations.

Step 4: Optimization problem. We maximize $S = x_{00} + x_{01} + x_{10} - x_{11}$ subject to PSD constraints (Gram matrix positive), submultiplicativity inequalities, and $|x_{ab}| \leq 1$.

By symmetry and convexity of the feasible set, the maximum occurs at

$$x_{00} = x_{01} = x_{10} = x, \quad x_{11} = -x.$$

This step is justified via convex extremization; details are given in Appendix A.

Step 5: Final bound. Substituting into constraints gives

$$(2 + \mu)x^2 \leq 1.$$

Thus

$$|S| = 4x \leq \frac{4}{\sqrt{2+\mu}}. \quad \square$$

5 Sectoral Coexistence

We provide an explicit kernel demonstrating that geometric consistency and Bell nonlocality can coexist in different sectors of a single kernel.

Let $\{v_i\}_{i \in \mathcal{G}}$ form a near-metric configuration (e.g., Euclidean embedding), let $\{w_j\}_{j \in \mathcal{B}}$ form a Tsirelson-optimal Bell configuration, and define cross-terms to be small: $\langle v_i, w_j \rangle = \epsilon$.

Then the combined Gram matrix remains PSD for sufficiently small ϵ , with the following properties:

- On $\mathcal{G}$: submultiplicativity holds with $\mu > 0$.
- On $\mathcal{B}$: CHSH = $2\sqrt{2}$ up to $O(\epsilon)$ corrections.

Thus a single kernel exhibits both behaviors.

6 Causality

We formalize directed kernels $K(i \rightarrow j)$ satisfying positivity, the composition inequality $K(i \rightarrow k) \geq \sum_j K(i \rightarrow j) K(j \rightarrow k)$, and acyclicity. Define causal order by

$$i \prec j \iff K(i \rightarrow j) > 0.$$

Theorem 6.1 (Independence). *Metric structure derived from G does not determine the causal order induced by K .*

Proof sketch. By explicit counterexample: we construct cases where identical G admits multiple incompatible K . $\square$

7 Discussion

The results of this paper admit a clean structural interpretation: geometry corresponds to transitivity of correlations, while nonlocality corresponds to the violation of transitivity.

Interpretation (heuristic). The parameter μ may be interpreted as a measure of compositional stability of correlations, analogous to contraction coefficients in data-processing inequalities [20] and to the role of metric constraints in the correlation polytope literature [8, 9]. The connection between $-\log |G|$ and approximate metric structure is grounded in Schoenberg’s classical theory linking positive definite kernels to metric embeddings [15, 19]. No physical mechanism is assumed.

8 Conclusion

We have established:

1. A rigorous tradeoff between geometric consistency and Bell nonlocality.
2. An intrinsic invariant $\mu(G)$ controlling this tradeoff.
3. Explicit coexistence constructions.

Open problems include: optimality of the bound $4/\sqrt{2+\mu}$, extension to multipartite scenarios, and physical realization of μ .

We emphasize that interpretations relating this framework to spacetime or physical systems remain speculative. See [17] for a comprehensive review of Bell nonlocality, and [5, 6, 18] for alternative principles bounding nonlocal correlations.

A Proof Details for Theorem 3.3

This appendix records a complete constrained-optimization route for the bound $|\text{CHSH}(G)| \leq 4/\sqrt{2+\mu}$.

A.1 Setup

Let $x_{ab} := G(A_a, B_b)$ for $a, b \in \{0, 1\}$, and let $S := x_{00} + x_{01} + x_{10} - x_{11}$. Since G is PSD and normalized, there exist unit vectors with $G(i, j) = \langle v_i, v_j \rangle$, so all Gram-matrix principal minors are nonnegative and $|x_{ab}| \leq 1$.

Define $y := G(B_0, B_1)$ and $z := G(A_0, A_1)$. From PSD for pairs and triples, one obtains (equivalently by Cauchy–Schwarz):

$$(x_{00} + x_{01})^2 \leq 2(1 + y), \quad (x_{10} - x_{11})^2 \leq 2(1 - y), \quad (1)$$

$$(x_{00} + x_{10})^2 \leq 2(1 + z), \quad (x_{01} - x_{11})^2 \leq 2(1 - z). \quad (2)$$

A.2 Incorporating geometric consistency

On the Bell sector $T = \{A_0, A_1, B_0, B_1\}$, assume

$$|G(i, k)| \geq \mu |G(i, j)| |G(j, k)|, \quad \mu \in [0, 1].$$

In particular,

$$|y| \geq \mu |x_{00}| |x_{01}|, \quad |y| \geq \mu |x_{10}| |x_{11}|, \quad (3)$$

$$|z| \geq \mu |x_{00}| |x_{10}|, \quad |z| \geq \mu |x_{01}| |x_{11}|. \quad (4)$$

These inequalities enforce a nontrivial lower bound on cross-setting consistency whenever the x_{ab} are large.

A.3 Reduction to symmetric extremal form

The feasible set (PSD constraints + submultiplicativity + box constraints) is compact. Hence a maximizer of $|S|$ exists.

By sign relabelings ($A_a \mapsto -A_a$, $B_b \mapsto -B_b$), permutation symmetry, and convex averaging over equivalent extremizers, we may reduce to the extremal ansatz

$$x_{00} = x_{01} = x_{10} = x, \quad x_{11} = -x,$$

with $x \geq 0$. Under this ansatz, $|S| = 4x$.

For this symmetric sector, the PSD constraints reduce to

$$2x^2 \leq 1 + |y|,$$

while submultiplicativity gives $|y| \geq \mu x^2$. Combining,

$$2x^2 \leq 1 + \mu x^2 \implies (2 + \mu)x^2 \leq 1.$$

Therefore

$$|S| = 4x \leq \frac{4}{\sqrt{2 + \mu}}.$$

This proves Theorem 3.3.

A.4 Consistency checks

- $\mu = 0$: $|\text{CHSH}| \leq 4/\sqrt{2} = 2\sqrt{2}$ (Tsirelson).
- $\mu > 0$: strict reduction below Tsirelson.
- Monotonicity: $C(\mu) = 4/\sqrt{2 + \mu}$ is strictly decreasing.

A.5 Remarks on rigor level

The only nontrivial reduction step is the symmetric-extremizer argument. In this draft, it is stated via standard convexity/symmetry reduction. A full formalization can be supplied either by:

1. a Lagrange/KKT derivation on the reduced feasible manifold, or
2. a semidefinite-program dual certificate showing the same extremal family is optimal.

B Coexistence Construction Details

We give a constructive family showing sectoral coexistence of approximate geometry and strong Bell nonlocality.

B.1 Block-Gram construction

Partition indices into geometric and Bell sectors: $S = S_G \sqcup S_B$. Choose unit-vector families

$$\{u_i\}_{i \in S_G} \subset \mathbb{R}^{d_G}, \quad \{w_j\}_{j \in S_B} \subset \mathbb{R}^{d_B},$$

with Gram blocks $G_G(i, i') = \langle u_i, u_{i'} \rangle$ and $G_B(j, j') = \langle w_j, w_{j'} \rangle$.

Assume G_G satisfies submultiplicativity on S_G with parameter $\mu_G > 0$, and selected $A_0, A_1, B_0, B_1 \in S_B$ realize a Bell-optimal pattern (Tsirelson-level in the $\mu = 0$ limit).

Embed both sectors orthogonally and add weak coupling ϵ :

$$\tilde{u}_i := (u_i, 0), \quad \tilde{w}_j := (\epsilon q_j, \sqrt{1 - \epsilon^2} w_j),$$

where $q_j \in \mathbb{R}^{d_G}$ are unit vectors. Then $G(i, j) = \langle \tilde{u}_i, \tilde{w}_j \rangle$ defines a normalized PSD kernel on all of S .

B.2 PSD and normalization

By construction, G is a Gram matrix of unit vectors, hence automatically PSD and normalized: $G(s, s) = 1$ for all $s \in S$. No additional PSD check is needed beyond explicit embedding.

B.3 Behavior on each sector

Geometric sector S_G . Restricting to S_G , the kernel is exactly G_G , so geometric consistency $\mu_G > 0$ is preserved.

Bell sector S_B . For $j, j' \in S_B$:

$$G(j, j') = (1 - \epsilon^2) \langle w_j, w_{j'} \rangle + \epsilon^2 \langle q_j, q_{j'} \rangle.$$

Hence Bell correlators are

$$E_{ab}(\epsilon) = (1 - \epsilon^2) E_{ab}^{(0)} + \epsilon^2 R_{ab},$$

with $|R_{ab}| \leq 1$. Therefore

$$\text{CHSH}(\epsilon) = (1 - \epsilon^2) \text{CHSH}^{(0)} + \epsilon^2 R, \quad |R| \leq 4,$$

so $\text{CHSH}(\epsilon) = \text{CHSH}^{(0)} + O(\epsilon^2)$. If $\text{CHSH}^{(0)} = 2\sqrt{2}$, then for sufficiently small ϵ , Bell nonlocality remains near-maximal.

B.4 Explicit finite-dimensional toy instance

Take $S_B = \{A_0, A_1, B_0, B_1\}$ and choose vectors in $\mathbb{R}^2$:

$$A_0 = (1, 0), \quad A_1 = (0, 1), \quad B_0 = \frac{1}{\sqrt{2}}(1, 1), \quad B_1 = \frac{1}{\sqrt{2}}(1, -1).$$

Then

$$E_{00} = E_{01} = E_{10} = \frac{1}{\sqrt{2}}, \quad E_{11} = -\frac{1}{\sqrt{2}},$$

so $\text{CHSH} = 2\sqrt{2}$. Combining this Bell block with any geometric block G_G using the ϵ -coupled embedding above, PSD is preserved automatically and coexistence follows.

B.5 Interpretation

This construction proves existence of kernels where one subset exhibits transitivity-compatible geometry ($\mu > 0$) and another subset exhibits strong Bell nonlocality. Thus the obstruction is global-uniform, not sector-local: a single relational substrate can host both behaviors provided constraints are not imposed uniformly across all triples.

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